

SIMATS ENGINEERING

CO3-ASSESSMENT TOOL 1

Experiments

COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

Dr.V.Parthipan

Code of Conduct:

I C KARTHIK REDDY(192425157) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
C KARTHIK REDDY

Criteria	Understanding of Concepts 2.5	Experimental Design 3	Use of Tools 2	Analysis of Results 1.5	Documentation 1	Total (10)
Q1	2.5	3	2	1	1	9.5
Q2	2	3	2	1	1	9
Q3	2.5	3	2	1	1	9.5
Q4	2.5	3	2	1	1	9.5

Faculty Signature

Experiments

QUESTIONS:4

Total Marks:40

1. Capture packets during a TCP connection setup. Identify the three packets involved in the handshake (SYN, SYN-ACK, and ACK). Demonstrate the role of each flag and how they establish a reliable connection. (BL4) (10)
2. Capture a DNS query using UDP in Wireshark. Compare the UDP header size with TCP and note the absence of sequence numbers or acknowledgments. Discuss how this lightweight design impacts speed and reliability. (BL4) (10)
3. Simulate packet loss (e.g., disconnect briefly) and capture traffic in Wireshark. Observe how TCP retransmits lost packets while UDP does not. Explain why this difference makes TCP reliable but slower compared to UDP. (BL4) (10)
4. Use Wireshark to capture DNS traffic while resolving a domain name. Identify the query type (A, AAAA, MX) and the response received. Measure the time taken for resolution and explain why DNS typically uses UDP.(BL4) (10)

Experiment 1: TCP Three-Way Handshake (SYN, SYN-ACK, ACK)

Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.flags.syn==1

No.	Time	Source	Destination	Protocol	Length	Info
113	1.843802	172.23.162.119	172.23.22.197	TCP	66	63035 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM=1
522	13.365581	172.23.22.197	4.150.223.107	TCP	66	50301 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM=1
558	13.613034	4.150.223.107	172.23.22.197	TCP	66	443 → 50301 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM=1
733	17.518788	172.25.180.135	172.23.22.197	TCP	66	52113 → 7680 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1
766	18.440156	172.25.180.135	172.23.22.197	TCP	66	[TCP Retransmission] [TCP Port numbers reused] 52113 → 7680 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1
800	20.269292	172.25.180.135	172.23.22.197	TCP	66	[TCP Retransmission] [TCP Port numbers reused] 52113 → 7680 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1

> Frame 113: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{1B45AF5E-6438-45A3-8C5D-665F59C7C677}, id 0

> Ethernet II, Src: 24:d5:e4:6d:bc:f7 (24:d5:e4:6d:bc:f7), Dst: IntelCor_90:88:b9 (80:38:fb:90:88:b9)

> Internet Protocol Version 4, Src: 172.23.162.119, Dst: 172.23.22.197

> Transmission Control Protocol, Src Port: 63035, Dst Port: 7680, Seq: 0, Len: 0

0000 00 38 fb 90 88 b9 24 d5 e4 6d bc f7 08 00 45 00 8\$E

0010 00 34 d6 a2 40 00 7f 06 13 b6 ac 17 a2 77 ac 17 4 .@.....w

0020 16 c5 f6 3b 1e 00 c4 35 7b 12 00 00 00 00 02 ..2...5 (2.....

0030 ff ff 0a 01 00 00 02 04 05 b4 01 03 03 08 01 01a.....

0040 04 02 ..

Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.flags.syn==1

113	1.843802					
522	13.365581					
558	13.613034					
733	17.518788					
766	18.440156					
800	20.269292					

> Frame 733: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{1B45AF5E-6438-45A3-8C5D-665F59C7C677}, id 0

> Ethernet II, Src: 24:d5:e4:6d:bc:f7 (24:d5:e4:6d:bc:f7), Dst: IntelCor_90:88:b9 (80:38:fb:90:88:b9)

> Internet Protocol Version 4, Src: 172.25.180.135, Dst: 172.23.22.197

> Transmission Control Protocol, Src Port: 52113, Dst Port: 7680, Seq=0, Win=64240, Len=0, MSS=1460, WS=256, SACK_PERM=1

0000 00 38 fb 90 88 b9 24 d5 e4 6d bc f7 08 00 45 00 8\$E

0010 00 34 d6 a2 40 00 7f 06 13 b6 ac 17 a2 77 ac 17 4 .@.....w

0020 16 c5 f6 3b 1e 00 c4 35 7b 12 00 00 00 00 02 ..2...5 (2.....

0030 fa f0 18 61 00 00 02 04 05 b4 01 03 03 08 01 01a.....

0040 04 02 ..

Wireshark - Packet 733 - Wi-Fi

> Frame 733: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{1B45AF5E-6438-45A3-8C5D-665F59C7C677}, id 0

> Ethernet II, Src: 24:d5:e4:6d:bc:f7 (24:d5:e4:6d:bc:f7), Dst: IntelCor_90:88:b9 (80:38:fb:90:88:b9)

> Internet Protocol Version 4, Src: 172.25.180.135, Dst: 172.23.22.197

> Transmission Control Protocol, Src Port: 52113, Dst Port: 7680, Seq=0, Win=64240, Len=0, MSS=1460, WS=256, SACK_PERM=1

0000 = Version: 4

0010 0101 = Header Length: 20 bytes (5)

0020 > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

0030 Total Length: 52

0040 Identification: 0xd045 (53317)

0050 > Flags: 0x40, Don't fragment

0060 ...0 0000 0000 0000 = Fragment Offset: 0

0070 Time to Live: 125

0080 Protocol: TCP (6)

0090 Header Checksum: 0x0a01 [validation disabled]

0100 [Header checksum status: Unverified]

0110 Source Address: 172.25.180.135

0120 00 38 fb 90 88 b9 24 d5 e4 6d bc f7 08 00 45 00 8\$E

0130 00 34 d6 a2 40 00 7f 06 13 b6 ac 17 a2 77 ac 17 4 .@.....w

0140 16 c5 f6 3b 1e 00 c4 35 7b 12 00 00 00 00 02 ..2...5 (2.....

0150 fa f0 18 61 00 00 02 04 05 b4 01 03 03 08 01 01a.....

0160 04 02 ..

Close Help

wireshark_Wi-Fi-F273.pcapng

Packets: 883 - Displayed: 6 (0.7%) - Dropped: 0 (0.0%)

Profile: Default

Experiment 2 — DNS Query Using UDP in Wireshark

The image displays two screenshots of the Wireshark network protocol analyzer interface, showing packet captures and their detailed views.

Top Screenshot: DNS Query Using UDP

- Packet List:** Shows a single packet (No. 276) at Time 6.150587, Source 192.168.0.239, Destination 255.255.255.255, Protocol DNS, Length 593. Info: Standard query 0x638e AAAA time.google.com.
- Packet Details:**
 - Header Checksum: 0x7919 [validation disabled] [Header checksum status: Unverified]
 - Source Address: 192.168.0.239
 - Destination Address: 255.255.255.255
 - User Datagram Protocol, Src Port: 53, Dst Port: 53
 - Source Port: 53
 - Destination Port: 53
 - Length: 41
 - Checksum: 0x3317 [unverified] [Checksum Status: Unverified]
 - [Stream index: 88]
 - [Timestamps]
 - UDP payload (33 bytes)
 - Domain Name System (query)
- Packet Bytes:** Shows the raw packet data in hexadecimal and ASCII. The ASCII part shows the domain name "time.google.com" and the query type "AAAA".

Bottom Screenshot: HTTP Packet

- Packet List:** Shows a single packet (No. 275) at Time 6.150587, Source 192.168.0.239, Destination 172.21.22.197, Protocol TCP, Length 66. Info: Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM=1.
- Packet Details:**
 - Frame 275: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{1B45AFE5-6438-45A3-8C5D-665F59C7C677}, id 0
 - Ethernet II, Src: 24:d5:e4:6d:bc:f7 (24:d5:e4:6d:bc:f7), Dst: IntelCor_90:88:b9 (80:38:fb:90:88:b9)
 - Internet Protocol Version 4, Src: 192.178.211.188, Dst: 172.23.22.197
 - 0100 = Version: 4
 - ... 0101 = Header Length: 20 bytes (5)
 - Differentiated Services Field: 0x00 (DSCP: CS4, ECN: Not-ECT)
 - Total Length: 52
 - Identification: 0xbe08 (48648)
 - Flags: 0x00
 - ... 0000 0000 0000 = Fragment Offset: 0
 - Time to Live: 122
 - Protocol: TCP (6)
 - Header Checksum: 0x2af0 [validation disabled] [Header checksum status: Unverified]
 - Source Address: 192.178.211.188
- Packet Bytes:** Shows the raw packet data in hexadecimal and ASCII. The ASCII part shows the domain name "time.google.com" and the query type "AAAA".

Experiment 3 — TCP Retransmission vs UDP Packet Loss

Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.analysis.retransmission

No.	Time	Source	Destination	Protocol	Length	Info
766	18.440156	172.25.180.135	172.23.22.197	TCP	66	[TCP Retransmission] [TCP Port numbers reused] 52113 → 7680 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1
880	20.269292	172.25.180.135	172.23.22.197	TCP	66	[TCP Retransmission] [TCP Port numbers reused] 52113 → 7680 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1

> Frame 766: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{1B45AF5E-6438-45A3-8C5D-665F59C7C677}, id 0

Ethernet II, Src: 24:d5:e4:6d:bc:f7 (24:d5:e4:6d:bc:f7), Dst: IntelCor_90:88:b9 (80:38:fb:90:88:b9)

Internet Protocol Version 4, Src: 172.25.180.135, Dst: 172.23.22.197

0100 = Version: 4

.... 0101 = Header Length: 20 bytes (5)

> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

Total Length: 52

Identification: 0xd046 (53318)

> Flags: 0x40, Don't fragment

...0 0000 0000 0000 = Fragment Offset: 0

Time to Live: 125

Protocol: TCP (6)

Header Checksum: 0x0a00 [validation disabled]

0000 80 38 fb 90 88 b9 24 d5 e4 6d bc f7 08 00 45 00 :8...\$.m...E

0010 00 34 d0 46 40 00 7d 06 0a 00 ac 19 b4 87 ac 17 :4f@.}......

0020 16 c5 cb 91 1e 00 66 87 e8 27 00 00 00 00 80 02 :-----f-'.-----

0030 fa f0 18 61 00 00 02 04 05 b4 01 03 03 08 01 01 :---a-'.-----

0040 04 02 ..

wireshark Wi-FISPF273.ocapno

Packets: 883 · Displayed: 2 (0.2%) · Dropped: 0 (0.0%)

Profile: Default

Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

dns udp

No.	Time	Source	Destination	Protocol	Length	Info
276	6.150587	192.168.0.239	255.255.255.255	DNS	593	Standard query 0x638e AAAA time.google.com

Wireshark · Packet 276 - Wi-Fi

...0 0000 0000 0000 = Fragment Offset: 0

Time to Live: 64

Protocol: UDP (17)

Header Checksum: 0x7919 [validation disabled]

[Header checksum status: Unverified]

Source Address: 192.168.0.239

Destination Address: 255.255.255.255

> User Datagram Protocol, Src Port: 53, Dst Port: 53

Source Port: 53

Destination Port: 53

Length: 41

Checksum: 0x3317 [unverified]

[Checksum Status: Unverified]

[Stream Index: 88]

> [Timestamp]

0000 ff ff ff ff ff ff e0 46 ee 38 96 d0 08 00 45 00 :-----F-8...E

0010 00 3d 00 00 40 00 40 11 79 19 c0 a8 00 ef ff ff :=@.@.y.....

0020 ff ff 00 35 00 35 00 29 33 17 63 8e 01 00 00 01 :...5.5.)3-.....

0030 00 00 00 00 00 04 74 69 6d 65 06 67 6f 6f 67 :-----t ime goog

0040 6c 65 03 63 6f 6d 00 00 1c 00 01 00 85 09 00 00 :le.com-'.-----

0050 00 00 00 00 00 00 00 00 00 00 ea 03 00 01 4c 00 :-----L-'.-----

0060 00 00 2a 00 00 00 05 c0 02 00 78 47 12 00 00 80 :--*-----xG...-

0070 27 00 32 00 00 00 6d 3f 81 16 4c 00 00 64 43 :'-2-...m?-L...dC

0080 81 16 4c 00 00 00 ff ff ff ff 00 00 00 00 00 00 :--L-----

0090 00 00 cd 02 f1 02 00 00 00 00 00 00 00 00 ac 89 :-----

00a0 01 00 00 00 7c a7 88 41 74 00 80 cc 9c 6c 73 e2 :----]-A t-----ls-

00b0 48 ad 9a 96 1e 23 24 d5 e4 6d bc f7 08 00 10 2c :H...\$\$.m-'.--.,

00c0 7d 00 00 20 00 00 00 00 aa aa 03 00 00 00 00 00 :}.-----

00d0 45 00 00 28 af 85 40 00 80 06 39 a5 ac 17 18 3b :E-(-.@-'.--9-...-1

00e0 17 c9 35 8a f0 12 01 bb 85 92 47 4e 09 58 2d 99 :\$S-----GN X-...-

00f0 50 10 00 ff a7 90 00 00 e3 e5 6e d1 8b 0b 60 a5 :P-----n-n-'.--..

0100 01 3f 00 00 05 20 00 00 00 00 00 00 00 01 00 :?-----

0110 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 :-----

No.: 276 · Time: 6.150587 · Source: 192.168.0.239 · Destination: 255.255.255.255 · Protocol: DNS · Length: 593 · Info: Standard query 0x638e AAAA time.google.com

Close Help

Experiment 4 — DNS Resolution in Wireshark

FileEditViewGoCaptureAnalyzeStatisticsTelephonyWirelessToolsHelp

idns

No.

Time

Source

Destination

Protocol

Length

Info

276

6.150587

192.168.0.239

255.255.255.255

DNS

593

Standard query 0x638e AAAA time.eoole.com

Wireshark - Packet 276 - Wi-Fi

> Frame 276: 593 bytes on wire (4744 bits), 593 bytes captured (4744 bits) on interface \Device\NPF_{1B45AFE5-6438-45A3-8C50-665F59C7C677}, id 0

> Ethernet II, Src: Netgear_38:96:d0 (e0:46:ee:38:96:d0), Dst: Broadcast (ff:ff:ff:ff:ff:ff)

> Internet Protocol Version 4, Src: 192.168.0.239, Dst: 255.255.255.255

> 0100 = Version: 4

> 0101 = Header Length: 20 bytes (5)

> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

> Total Length: 61

> Identification: 0x0000 (0)

> Flags: 0x40, Don't fragment

> ...0 0000 0000 0000 = Fragment Offset: 0

> Time to Live: 64

> Protocol: UDP (17)

> Header Checksum: 0x7919 [validation disabled]

> [Header checksum status: Unverified]

> Source Address: 192.168.0.239

> Frame 276: 593 bytes on wire (4744 bits), 593 bytes captured (4744 bits) on interface \Device\NPF_{1B45AFE5-6438-45A3-8C50-665F59C7C677}, id 0

> Ethernet II, Src: Netgear_38:96:d0 (e0:46:ee:38:96:d0), Dst: Broadcast (ff:ff:ff:ff:ff:ff)

> Internet Protocol Version 4, Src: 192.168.0.239, Dst: 255.255.255.255

> 0100 = Version: 4

> 0101 = Header Length: 20 bytes (5)

> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

> Total Length: 61

> Identification: 0x0000 (0)

> Flags: 0x40, Don't fragment

> ...0 0000 0000 0000 = Fragment Offset: 0

> Time to Live: 64

> Protocol: UDP (17)

> Header Checksum: 0x7919 [validation disabled]

> [Header checksum status: Unverified]

> Source Address: 192.168.0.239

0000 ff ff ff ff ff ff ff ff ee 38 96 d0 00 00 45 00F.8....E.

0010 00 3d 00 00 40 00 11 79 19 c0 a8 00 ef ff ff@.y.....

0020 ff ff 00 35 00 35 00 29 33 17 63 8e 01 00 00 01 ...5.5.)3c.....

0030 00 00 00 00 00 04 74 69 6d 65 06 67 6f 6f 67t ime goog

0040 6c 65 03 63 6f 6d 00 00 1c 00 01 00 85 09 00 00 le.com.....

0050 00 00 00 00 00 00 00 00 00 ea 03 00 01 4c 00L.....

0060 00 00 2a 00 00 00 05 c0 02 00 78 47 12 00 00 00 ..8.....x0....

0070 27 00 32 00 00 00 6d 3f 81 16 4c 00 00 00 64 43 '2...m? ..L...dC

0080 81 16 4c 00 00 00 ff ff ff ff 00 00 00 00 00 00 ..L.....

0090 00 00 cd 02 f1 02 00 00 00 00 00 00 00 00 ac 89[.A t...ls

00a0 01 00 00 00 7c a7 88 41 74 00 80 cc 9c 6c 73 e2H...#5...a...;

00b0 48 ad 9a 96 1e 23 24 d5 e4 6d bc f7 60 00 10 2cGN X...

00c0 17 c9 35 8a f0 12 01 bb 85 92 47 4e 09 58 2d 99 ...5.....n....

00d0 50 10 00 ff a7 90 00 00 e3 e5 6e d1 80 00 60 a5 P.....?

00e0 01 3f 00 00 05 20 00 00 00 00 00 00 00 01 00?

00f0 00 80 00 00 00 00 00 00 00 00 00 00 00 00 00?

0100 00 80 00 00 00 00 00 00 00 00 00 00 00 00 00?

0110 00 80 00 00 00 00 00 00 00 00 00 00 00 00 00?

Domain Name System: Protocol

Packets: 883 · Displayed: 1 (0.1%) · Dropped: 0 (0.0%)

Profile: Default